



Practice Problem Set -01 (Successive differentiation)

Q.01 If $y = x \log \frac{x-1}{x+1}$, show that $y_n = (-1)^{n-2} (n-2)! \left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n} \right]$.

Q. 02: Find the n th derivative of

(i) $\cos x \cos 2x \cos 3x$

(ii) $e^{2x} \cos^2 x \sin x$.

(iii) $\cos 2x \cos 3x$

(iv) $\sin^3 x \cos^2 x$

(v) $e^{5x} \cos x \cos 3x$

(vi) $\frac{x^2}{2x^2+7x+6}$

(vii) $\frac{x}{x^2+a^2}$